

mHealth Platform

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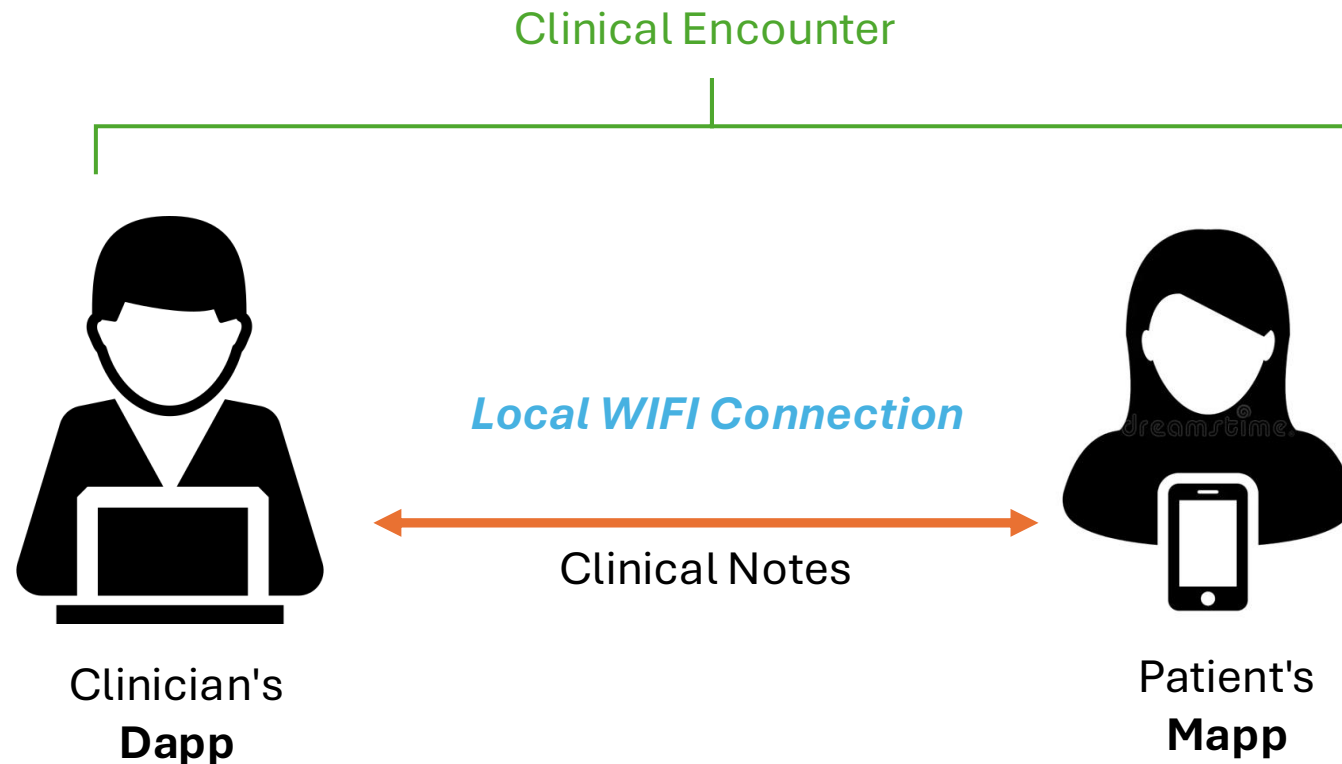
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Motivation

- Implement EMR solution to replace the current paper-based system where handwritten clinical notes are provided to the patients
 - Clinical notes can be lost
 - Clinical notes are cumbersome to carry
 - Can be difficult to read after a few years
- Infrastructure constrains
 - Cloud access is rare or not available
 - Internet can be spotty
 - Limited compute resources

Solution

- Leverage Desktop/Labtop computers available in local dispensaries/hospital
- Leverage the increase adoption of smart phones



Dapp: Features

- Note summarization and synopsis using LLM
- Vitals intake form with OCR/handwriting recognition
- Recurring actions with mobile phone notification and patient input
- Referral letter with patient clinical history (LLM)
- Additional document attachment with OCR/handwriting recognition
- Clinical keywords dictionary and clinical keywords extraction
- Patient data backup and recovery

Dapp: you should also know...

- The current version includes 'mocked data'
- To enable access to all the functionalities several applications need to be installed:
 - Ollama: LLM for *handwriting to txt* and *notes summarization*
 - Tesseract: *jpeg processing/OCR*
 - Poppler: *converting pdf to txt*
- a local model (qwen2.5vl:3b) needs to be downloaded for Ollama
 - qwen2.5vl:3b is 3.2GB
 - can be stored on a USB stick to be redistributed

Validation and Verification!

- The note summarization feature needs additional validation
- The vital form intake needs verification after each scan
- More complex LLM models could be substituted but comes at the cost of:
 - disk/memory space
 - performance (stronger CPU/GPU)